

Huffman Coding Greedy Method

Made with

KINEMASTER
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⇒ ① Compression technique.

② Used for reducing the size of data or message.

1. Fixed length code

2. Variable length code

Message → B B B C A A A D E E E A B B B A C C D A → 20

$$20 \times 8 = 160 \text{ bits}$$

<u>char</u>	<u>ASCII</u>	<u>code</u>
A	65	01000001
B	66	01000010
C	67	01000011
D	68	01000100
E	69	01000101

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2. Variable length code.

Fixed Length Code:

B B B C A A A D E E E A B B B A C C D A
 000 000 001 010 - - - - -

Character	count/frequency	Code
A	6	000
B	6	001
C	3	010
D	2	011
E	3	100

$5 \times 8 = 40 \text{ bits}$

$5 \times 3 = 15 \text{ bits}$

$+ = 55 \text{ bits}$

$2^8 = 256$

$2^n = 2^2 = 4$
 $2^3 = 8 (10-7)$

$20 \times 3 = 60 \text{ bits (memory)}$

$60 + 55 = 115 \text{ bits}$
 reduce

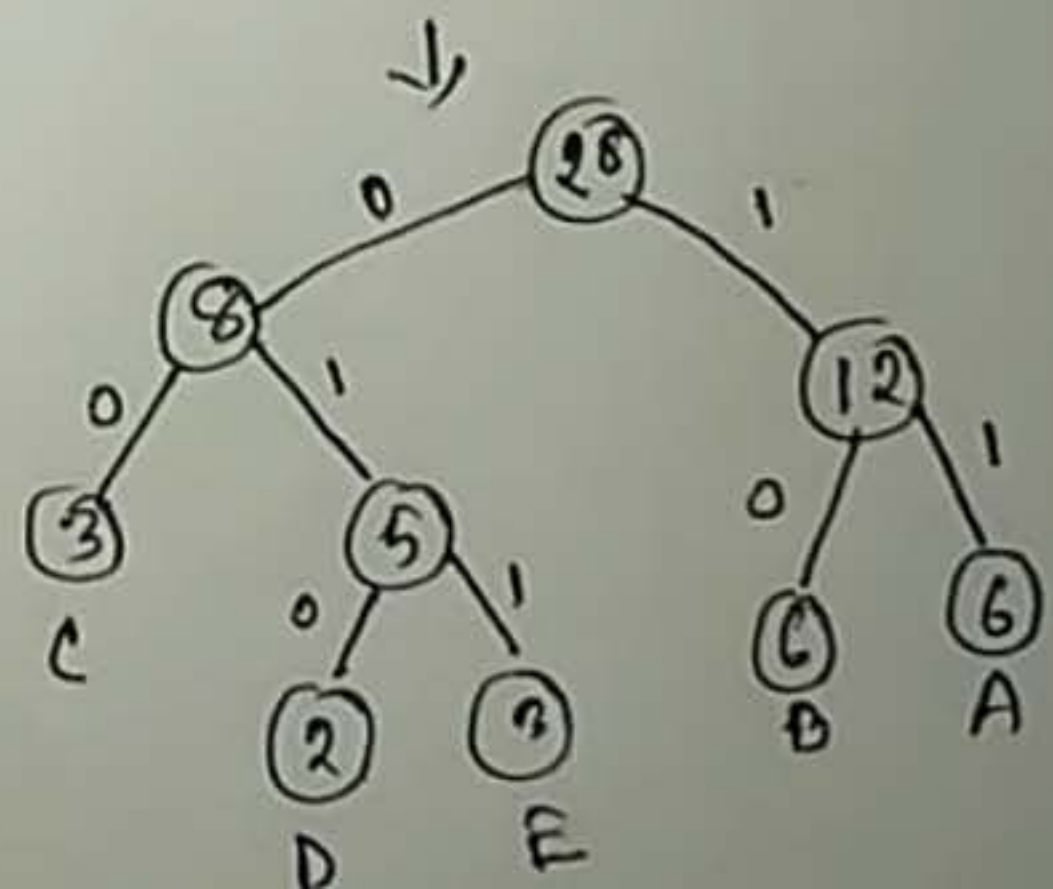
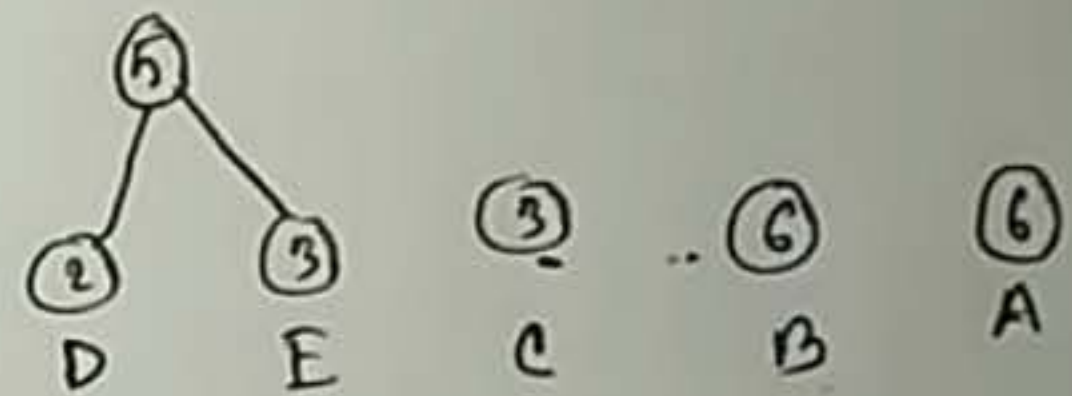
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Char	ASCII
A	65
B	66
C	67
D	68
E	69

Variable Length Code:

character	count/frequency	Code	
A	6	11	$6 \times 2 = 12$
B	6	10	$6 \times 2 = 12$
C	3	00	$3 \times 2 = 6$
D	2	010	$2 \times 3 = 6$
E	3	011	$3 \times 3 = 9$
40 bits		12 bits	(45 bits)

B B B C A A A A
10 10 10 00 1111



52 bits

$45 + 52 = 97 \text{ bits}$
reduced